

What we did, in order, and why

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Where we are

We are asking whether a network that sees the world through a log-polar, fixation-driven front end shows the face inversion effect the way people do. Two human experiments set the bar: Yin (1969), a memory task, and Dobs/Kanwisher (2023), a matching task. A model is doing well if faces cost a lot when you turn them upside down and objects and houses cost almost nothing.

Four things now work:

- **The matching Kanwisher task is matched by all four models.** On our best face set, r7, r8, r9 and David's model all land on the human 87.50 upright accuracy with an inversion effect of +10.8 to +12.2, against a human +10.70 to +11.60.
- **The memory Yin task is matched too,** on a face set that is three-quarters familiar identities: 95.75 upright and a +12.17 cost, where the humans got 96.29 and +12.16.
- **Objects behave like the control they should be.** Their inversion cost is near zero in every run, which is what people show.
- **We have a lead on the houses, in one model.** Swapping known buildings for unknown ones in the test set steadily raises David's model's house inversion cost, from +0.4 all-known to +10.0 all-unknown, passing the human +2.63 on the way. Run on the same sets, r8 and r9 stay flat at +9 to +11 — so this is something that model's training gives it, not something the architecture does.
- **Ordering still stands.** The Yin inversion cost goes faces > houses > objects in every model, which is the order people show.

All four orientation cells, at the one noise level per model from Step 13, so the three categories are comparable within a table. 100 seeds each. UU–UI is the cost Yin reports; UU–II is there because it tells a different story.

David's model, $p = 0.26$

	UU	II	UI	IU	UU–UI	UU–II
Faces	89.25 $\pm$ 0.62	82.33 $\pm$ 0.82	75.25 $\pm$ 0.82	76.12 $\pm$ 0.82	+14.00	+6.92
Houses	90.04 $\pm$ 0.52	88.96 $\pm$ 0.57	81.67 $\pm$ 0.70	78.92 $\pm$ 0.70	+8.38	+1.08
Objects	98.83 $\pm$ 0.21	98.42 $\pm$ 0.24	95.83 $\pm$ 0.39	96.25 $\pm$ 0.29	+3.00	+0.42

r7, $p = 0.30$

	UU	II	UI	IU	UU–UI	UU–II
Faces	86.25 $\pm$ 0.61	80.54 $\pm$ 0.81	74.21 $\pm$ 0.90	74.96 $\pm$ 0.78	+12.04	+5.71
Houses	91.38 $\pm$ 0.50	83.96 $\pm$ 0.59	81.00 $\pm$ 0.62	82.96 $\pm$ 0.62	+10.38	+7.42
Objects	98.04 $\pm$ 0.25	97.71 $\pm$ 0.28	95.58 $\pm$ 0.34	96.21 $\pm$ 0.34	+2.46	+0.33

r8, $p = 0.00$

	UU	II	UI	IU	UU–UI	UU–II
Faces	92.12 $\pm$ 0.53	90.96 $\pm$ 0.53	79.50 $\pm$ 0.82	79.13 $\pm$ 0.82	+12.62	+1.17
Houses	84.83 $\pm$ 0.60	81.46 $\pm$ 0.64	73.87 $\pm$ 0.71	74.33 $\pm$ 0.75	+10.96	+3.38
Objects	98.29 $\pm$ 0.25	99.17 $\pm$ 0.18	97.12 $\pm$ 0.28	97.00 $\pm$ 0.32	+1.17	–0.88

Humans (Yin 1969)

	UU	II	UI	IU	UU–UI	UU–II
Faces	96.29	81.88	84.13	78.58	+12.16	+14.41
Houses	90.71	85.75	88.08	85.71	+2.63	+4.96
Objects	84.79	83.96	86.71	82.75	–1.92	+0.83

The two mismatch cells, UI and IU, land within about a point of each other in every row, so what the models pay is a cost for study and test orientation disagreeing, not a cost for seeing something upside down. Measuring with UU–II instead would say something quite different — David’s houses would read +1.08 against a human +4.96, and r8’s faces +1.17 against +14.41 — which is why we use UU–UI throughout, as Yin does.

The one thing still open: no house set matches human accuracy and human inversion cost at the same time. The rungs that give the right cost are ones where the model is at 98% accuracy and the humans were at 91%.

The rest of this document is the story of how we got here, with the numbers.

The yardstick

"Good" means **the model behaves like a human on the inversion effect** — not that it is accurate. Two human experiments:

	Faces	Objects	Houses
Yin (1969) upright-study/upright-test accuracy	96.29	84.79	90.71
Yin inversion cost (upright-upright – upright-inverted)	+12.16	–1.92	+2.63
Dobs/Kanwisher (2023) upright matching accuracy	87.50	—	—
Kanwisher inversion effect	+10.70 (between-subj) / +11.60 (within-subj)	—	—

So the target is a **big face cost and near-zero object/house cost**, at accuracies close to the human ones. Two knobs let us line a model up with a human: which face pictures we test on, and one retrieval-noise parameter p (bit-flip noise on the 256-d code) that is tuned until upright accuracy sits on the human anchor. Everything below is a fight over those two knobs.

Step 1 — The models we use

Model	Faces	Objects	Houses	Total classes
r7	128	64	201 ZuBuD — all of them	393
r8	128	64	40 ZuBuD	232
r9	128	64	40 ZuBuD	232
David's (<code>house_control_r1</code>)	128	128	40 ZuBuD	296

All four models train on ZuBuD buildings; three of them on a 40-building subset. David's house store is `object0101.view02`-style ZuBuD files — 40 buildings $\times$ 3 views = 120 training images — i.e. **the same dataset and the same class count as r8 and r9**.

r7 grows all three categories on one shared schedule ($4 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow$ all classes, 60 epochs).

r8 grows each category on *its own* schedule — objects saturate early, faces keep growing, houses arrive late and stay few — and caps houses at 40:

Stage	1	2	3	4	5	6
Faces	4	8	16	32	64	128
Objects	4	16	32	64	64	64
Houses	0	0	4	8	16	40
Epochs	6	6	8	8	10	22

r9 is r8 with one thing changed: a balanced diet. Objects supply $\sim 1,000$ images per class and houses only 3, so the natural diet is $\sim 90\%$ object images. r9 forces every batch to be faces 45% / objects 45% / houses 10%.

David's model (`house_control_r1`) **is a curriculum model**, on a two-stage schedule rather than r8's six-stage one. Its training history records:

Stage	stage_spec	Active classes	Epochs
1	all-prehouse	256 (128 faces + 128 objects)	2
2	all	296 (houses added)	29

So houses are withheld and then introduced, which is the developmental move; what it does *not* do is grow the face and object class counts (all 128 of each are present from epoch 1). The `curriculum: false` in its `config.json` refers only to that class-count ladder — the house delay runs off a separate `houses_delay_epochs: 2` setting, which is why there are no `stage*.pth` checkpoints. Alongside it, category proportions are held fixed at faces 40% / objects 40% / houses 20% for the whole run.

Two things still differ from r8 and matter when reading its rows:

1. **128 object classes**, not 64.

2. **No learning-rate schedule**, where r7/r8/r9 use one global cosine decay.

So r8 vs David's model is a *curriculum-shape + object-count + LR-schedule* contrast, not a single-variable one. For a clean single-variable house contrast use r8 vs r9.

Accuracy on held-out upright images:

Model	Overall	Faces	Objects	Houses
r7	79.5	92.1	75.8	96.0
r8	79.3	91.0	76.3	90.0
r9	78.8	91.1	75.5	92.5
David's	57.8	91.5	53.4	97.5

David's overall number is low because it carries 128 object classes and trained without an LR schedule. Its *faces* are as good as everyone else's, which is why it stays usable for the face simulations.

Step 2 — first simulations, on stimuli the models trained on

We ran both simulations on the 128 CelebA identities, the 64 object classes and the ZuBuD houses — all things the model had seen — with 200 seeds, calibrating p separately per category.

Yin, r7, 200 seeds (runs/sim_seeds/results_r7_curriculum.csv):

Category	p	UU	II	UI	IU	Cost (UU–UI)	Human cost
Faces (128, trained)	0.38	93.10 ± 0.29	81.00 ± 0.51	81.54 ± 0.49	81.90 ± 0.49	+11.56 ± 0.55	+12.16
Houses (201 ZuBuD, trained)	0.32	90.29 ± 0.40	82.00 ± 0.46	79.08 ± 0.50	81.33 ± 0.51	+11.21 ± 0.47	+2.63
Objects (64, trained)	0.43	72.62 ± 0.59	69.29 ± 0.68	71.21 ± 0.65	70.48 ± 0.71	+1.42 ± 0.81	–1.92

Kanwisher, r7, 200 seeds. Dobs et al. Exp. 5 tested faces only, so faces are the only row with a human reference:

Category	p	Upright	Inverted	Effect	Human
Faces (128, trained)	0.43	85.44 ± 0.10	70.81 ± 0.13	+14.63 ± 0.13	87.50 / +10.70–11.60

Two things to notice, one of which we did not notice at the time:

- **Faces and objects looked right.** The Yin face cost (+11.56) sits within one SEM of the human +12.16, and objects are near-flat in both simulations (+1.42 under Yin).
- **Houses were already wrong here.** The Yin house cost is **+11.21 against a human +2.63** — the failure Step 12 is about was present in the very first sweep, on trained buildings, and it never went away. It is also already *specific to Yin*: the same model on the same buildings shows almost no house effect under Kanwisher. The Yin-vs-Kanwisher dissociation that is the project's open question was there from the first result.

Also: this stage was r7 only. r8, r9 and David’s model did not exist yet, and every face in it was an identity the network had been trained to classify.

Step 3 — the problem: our test faces were the wrong faces

Yin’s participants recognised **unfamiliar adult male faces**. Dobs/Kanwisher’s participants matched **female celebrity faces** they may or may not have known. We were testing on 128 identities the network had been explicitly trained to classify — so the model could win by looking up a classifier unit it already owned, not by remembering or perceiving anything. Same problem for houses: the model was being tested on buildings it had trained on.

Step 4 — the fix: bring in the real stimulus sets

We packed the two Dobs/Kanwisher stimulus sets as held-out probes. Neither contributes a single gradient step to any model.

- **Set A** — 40 identities $\times$ 5 photos = 200 photographs. Every identity is a **young white female celebrity**; the set is deliberately homogeneous.
- **Set E** — 64 identities $\times$ 10 photos = **640 photographs**, anonymised `id001–id064`. Verified by eye: **both sexes, a wide age range, varied ethnicity**, and unconstrained in-the-wild photography (press shots, sports, studio, snapshots). It is a far more *heterogeneous* pool than Set A in every dimension, not just race — that is what makes identities in it easier to tell apart.

For houses we started holding buildings out: r8, r9 and David’s model are each scored on the 161 ZuBuD buildings they never trained on, and later we built `houses_common` — 94 buildings from a different house dataset that none of the models ever saw.

Step 5 — the first Yin design was broken, and how we found out

Before any of the results below, there is a design failure worth putting on the record, because it produced the best-looking face numbers in the project and all of them were false.

The design. Yin’s task needs 40 study identities plus 24 *never-studied* distractors — 64 identities in total. Set A has exactly 40. So the first version studied **all 40 Set A identities** and drew the **24 distractors from the trained faces store**, i.e. from celebrities the network had been trained to classify.

It worked beautifully. Upright-upright accuracy across the three models and every noise level we tried:

Model	p=0.00–0.10	p=0.32	p=0.34	p=0.36	p=0.38
r8	95.42 $\pm$ 0.50	91.25	90.21	89.38	87.71
r9	93.75 $\pm$ 0.68	89.79	89.58	86.88	85.83
David’s	97.08 $\pm$ 0.50	95.42	94.37	92.92	91.04

Near the 96.29 human anchor, on genuinely novel faces. It was, briefly, the result we wanted.

Three things gave it away.

1. **Noise barely moved it.** Upright accuracy sat flat near 95.8% across the whole noise range before finally sagging — a calibration knob that does not bite means the task is not being solved the way we think it is.
2. **Unstudied items scored as well as studied ones.** Set A items the model had *never been shown in the study phase* still beat old-store distractors ~90% of the time. Nothing about memory can explain that.
3. **The same faces scored at chance in an honest test.** Set A against Set A distractors: **53.33%**.

The diagnosis. The two answer options came from two different photo collections, so the model could win by spotting **which photo set a picture came from** — "which of these two looks like it came from the pile I was just shown?" — with no memory involved. The 40-point gap between the two ways of scoring is how big that shortcut was.

We threw all of those runs away. The rule we took from it: the studied items and the distractors have to come from the same set of photos.

The fix: build mixed stores by ratio

The rule creates a problem — Set A alone cannot fill a 64-identity design. The answer was to stop treating "familiar" and "unfamiliar" as separate *stores* and start treating them as a *composition within one store*: `make_faces_mix64.py` packs a single store from any mix of groups, tagging each identity `celeb_`, `known_`, `setE_` OR `setA_`, so the study and distractor pools are drawn from one pool while the known/unknown ratio is whatever we choose. Everything after this point — `faces_mixed64`, the mixu ladder, the familiarity sweeps — is built that way.

It also lets us **test for the shortcut directly**, by building the same design twice with familiarity pointing opposite ways ($r8$, $p=0.36$, 100 seeds):

Store	Studied	Distractors	UU	Cost
<code>faces_k40u24</code>	40 known	24 unknown	96.50 $\pm$ 0.34	+4.17 $\pm$ 0.61
<code>faces_u40k24</code>	40 unknown	24 known	63.92 $\pm$ 0.88	+25.04 $\pm$ 1.08

Same identities, same design, same noise — only which side familiarity favours is flipped, and accuracy swings **33 points**. Note that `faces_u40k24` is the honest single-store version of exactly the broken design above (unknown studied, known distractors): it reads **63.92**, where the cross-store version read 95.42. That gap *is* the shortcut, measured.

Neither of these is a real result — a two-choice test that can be won by spotting a face the model was trained on is not measuring memory, in either direction. They are a check, and we now run that check every time familiarity lines up with the right answer.

Step 6 — Set A does not work. At all.

This is the big negative result and it cost a lot of runs.

Kanwisher on Set A (design is sound — one store throughout — the numbers are simply bad):

Model	p	Upright (human 87.50)	Effect (human +10.70)
r8	0.00	71.25 $\pm$ 0.02	+2.90 $\pm$ 0.03
r9	0.00	69.86 $\pm$ 0.02	+3.41 $\pm$ 0.03
David's	0.00	71.18 $\pm$ 0.03	+2.69 $\pm$ 0.03

Upright tops out ~16 points below the human anchor **at zero noise**, so p has nowhere to go — the calibration knob is dead. The inversion effect is a quarter of the human one. Adding noise only makes it worse (~60% at $p \approx 0.4$).

Yin on Set A is at chance in an honest same-store test: 48.9–68.9% upright-upright across five models, against a 96.29 anchor.

Why Set A fails: its photographs of one person barely resemble each other (within-identity pixel correlation **+0.078** vs Set E's **+0.113**), while every identity is a young white female celebrity, so between-identity variability is low too. Both directions hurt an identity-matching task. It is *not* a resolution problem — Set E actually has more sub-224px images.

That 53.33% is also what condemned the cross-store design in Step 5: the same identities read 95.42 when their distractors came from another photo collection.

Set E, by contrast, worked: r8 96.29 / +23.67, r9 92.58 / +24.00, David's 96.83 / +18.46. Accuracy lands on the anchor, but the cost is 1.5–2 $\times$ the human +12.16 — unfamiliar-only faces *overshoot*.

Objects: accuracy too high, inversion effect right

Worth stating once here, because it holds in every run from this point on and is easy to misread as a failure. Objects are **always** far above their human accuracy anchor — r8 reads 98.29 upright-upright against Yin's 84.79, and the other models 98–99 — for an unavoidable reason: **every object class was trained on**, no object item is ever held out, and no noise level brings them down without collapsing faces at the same p .

But the number that matters is right. The object inversion cost is **+1.17 (r8)**, **+1.67 (David's)**, **+1.42 (r7)** against a human -1.92 , and $\overline{u-u}$ is **+0.42** against a human **+0.83** — i.e. **near-zero and flat, which is exactly the human pattern**. So objects do the job we need them to do — the category that should *not* show an inversion effect — they just do it from a ceiling. When I quote objects anywhere below, the number that means something is the cost, not the accuracy.

Step 7 — the familiar half: 24 celebs the models actually trained on

Reasoning: Yin's and Kanwisher's participants are not blank slates. Some test identities should be familiar. We picked 24 similar-looking trained CelebA identities (Adele ... Tilda Swinton), using photos from the held-out `valid` split so the *identity* was seen but the *image* was not.

24 items cannot fill a 40/24 design, so Yin scales down to 15 study / 9 test (the same 5:3 ratio). Nine test pairs = 11.1 points per pair, so read the SEM.

The noise-calibration bug : `p` was fitted on one set of item draws (base seed 42) and reported on a *different* set (seeds 101–120) — picking the noise level on one set of draws and reporting it on another, which was worth 1–2.6 points on a 24-identity set. After the fix (`-calib-on-eval-seeds`), 100 seeds at `p=0.30`:

Model	Faces UU (96.29)	Faces cost (+12.16)	Houses at same p	Objects at same p
r8	96.33 ±0.61	+7.00 ±0.85	75.54 / +8.46	95.75 / +1.54
David's	98.56 ±0.41	+10.55 ±1.10	86.96 / +8.04	98.29 / +3.04

"Houses work well in David's model" means relative **accuracy** (86.96 vs 75.54), is better but its house inversion **cost** is just as wrong as everyone else's (+8 against a human +2.63). Accuracy fixed, cost not fixed.

Step 8 — mixing 24 celebs with 40 Set A faces: results fall again

`faces_mix64` = 24 trained celebs + all 40 Set A identities, 64 items, 20 seeds:

Model	p	UU (96.29)	Cost (+12.16)
r8	0.00	83.96 ±1.29	+11.46 ±1.28
David's	0.03	84.17 ±1.44	+13.96 ±2.01

`p` pinned at the floor for every model: 40 of the 64 items carry no signal, so no noise level reaches 96.29. The cost looks reasonable but it is 24 working items diluted by 40 dead ones. **Not reportable as a calibrated Yin result.**

Step 9 — Kanwisher on the same mix: pretty good

This was much better than Yin, Kanwisher lands on its 87.5 anchor exactly, because the task has no memory phase and the anchor is reachable. Identity pool grown 24 → 49 by adding Set A identities:

Identity pool	24	29	34	39	44	49
r8 upright	95.81	93.59	91.37	89.49	87.99	86.77
r8 effect	+6.08	+5.34	+5.06	+4.76	+4.59	+4.44
David's upright	96.01	94.25	92.03	90.29	88.63	87.34
David's effect	+6.26	+5.76	+5.45	+5.14	+5.03	+4.90

Accuracy passes through the human anchor, but the **inversion effect is only about half the human +10.70** everywhere. A second sweep holding the pool at 40 and swapping celebs for Set A one at a time gave the same steady fall (+4.9 → +2.7 as the last celeb leaves).

Both sweeps ran near $p \approx 0$. Refitted where the anchor actually binds ($p \approx 0.30$ – 0.33) the effects grow by about a third and the models finally separate: r8 **+6.11 ± 0.23**, David's **+7.84 ± 0.28**. Still below human, but now outside the error bars, and in the same order as the Yin celeb-24 run.

Caveat on this whole sweep: it cannot test *familiarity*, because all models trained on the same 24 celebs. What it varies is *which stimulus set* the pool is drawn from, and Set A is simply harder.

Step 10 — the mixes that work: `faces_mixed64` and the familiarity ladder

Rather than keep guessing compositions, we did two things at once: settled on one 64-identity mix for the headline runs, and built a **ladder** — one 64-identity store per known/unknown ratio, so the Yin design is *identical* at every rung and only the ratio moves. The two belong together: the headline store is one rung of the ladder, and the ladder is what tells you whether that rung was lucky.

faces_mixed64 = 24 named celebs + 16 other trained identities + 12 Set E + 12 Set A — 64 identities
 × 5 photos: 40 known, 24 unknown.

Kanwisher on `faces_mixed64` — all four models replicate both human numbers

Model	p	Upright (87.50)	Effect (+10.70 / +11.60)
r7	0.40	87.33 ± 0.30	+11.09 ± 0.30
r8	0.40	87.13 ± 0.32	+10.79 ± 0.38
r9	0.40	87.08 ± 0.36	+11.15 ± 0.38
David's	0.39	87.33 ± 0.35	+12.17 ± 0.26

All four land on the 87.50 upright anchor and inside (or within half a point of) the two human effects. **This is the strongest face result in the project.**

Yin on the ladder — and where `faces_mixed64` sits on it

r8, one fixed p=0.23, 100 seeds per rung:

Unknown identities	8	16	24 (= mixed64)	32	40	48
Upright-upright	95.96	95.75	89.63	87.83	85.67	81.75
Inversion cost	+9.04	+12.17	+12.21	+10.83	+11.79	+10.33

Two things fall out:

- `faces_mixed64` **gets the cost right (+12.21 vs human +12.16) but its accuracy is 7 points low**. It is not a lucky rung — the whole 16–40 region gives a human-sized cost — but it is not the accuracy optimum either.
- **The best Yin faces result in the project is one rung over: `faces_mixu16` (48 known + 16 unknown), r8, p=0.23 → 95.75 ±0.42 upright-upright (human 96.29) and +12.17 ±0.65 cost (human +12.16).** Both anchors at once, which nothing else in the project manages.

So the honest summary is: **Kanwisher is matched on `faces_mixed64`; Yin is matched one rung more familiar, on `faces_mixu16`**. Accuracy falls steadily as the faces get less familiar, as you would expect; the cost does not — it peaks around 16–24 unknown and flattens.

For completeness, the same four compositions at a matched p=0.30 across all three models:

Store	r8 UU / cost	r9 UU / cost	David's UU / cost
64 known	95.00 / +6.92	95.46 / +10.46	97.83 / +9.12
32 known + 32 unknown	86.37 / +12.00	85.17 / +12.83	90.42 / +16.00
40 known + 24 unknown (mixed64)	87.17 / +12.00	86.96 / +12.87	87.08 / +14.21
64 unknown	88.29 / +15.92	83.58 / +14.50	87.12 / +17.50

The two simulations disagree about familiarity

Read the last table against Step 9 and the contradiction is plain:

- **Under Yin, the inversion cost grows as faces get less familiar** (r8: +6.9 all-known → +15.9 all-unknown).
- **Under Kanwisher, the inversion effect grows as faces get *more* familiar** (r8: +6.1 at 24 celebs → +2.8 at zero celebs).

One thing to note about every set on this ladder: none of them lets familiarity sit on the same side as the right answer. We shuffle the identities before splitting them into studied and not-studied, so known and unknown faces land on both sides. Step 5 is what taught us to do that.

Step 11 — what r9 (the balanced diet) adds

r9 is r8 with the diet rebalanced to faces 45 / objects 45 / houses 10, and nothing else. It is the only place where a single change can be credited with a single effect, and it does three things:

Measure (Set E stimuli, 100 seeds)	r8	r9	Human
Yin houses, held-out ZuBuD, upright-upright	84.83	92.00	90.71
Yin houses, cost	+10.96	+10.75	+2.63
Yin faces, upright-upright	96.29	92.58	96.29
Yin faces, cost	+23.67	+24.00	+12.16
Yin objects, cost	+1.17	+4.58	−1.92
Kanwisher faces (Set E), upright / effect	87.76 / +6.45	88.98 / +5.92	87.50 / +10.70
Kanwisher faces (faces_mixed64), upright / effect	87.13 / +10.79	87.08 / +11.15	87.50 / +10.70–11.60

1. **The diet fixes house accuracy.** r9 is the *only ZuBuD-trained model that reaches the human house accuracy anchor* (92.00 vs 90.71) — r8 with the same 40 house classes and the natural diet sits 6 points lower. Giving houses a guaranteed 10% of each batch is worth ~7 points on buildings it never trained on.
2. **It does not touch the house inversion cost** (+10.75 vs +10.96). More house exposure buys accuracy, not human-like orientation tolerance. That is direct evidence the house problem is not a data-starvation problem.
3. **It costs objects.** r9's object inversion cost triples (+1.17 → +4.58), moving away from the human −1.92, and face accuracy drops ~4 points.
4. **On the headline face store it is indistinguishable from r8.** Run on faces_mixed64 at the same $p=0.40$, r9 reads **87.08 ± 0.36 upright and +11.15 ± 0.38** against r8's 87.13 / +10.79 — both on the human anchors, well inside each other's error bars. The balanced diet changes houses and objects; it does not change the face result.

r9 also gives the best Yin cost match on faces_mixed64 at $p=0.30$ (+12.87 vs human +12.16, against r8's +12.00 and David's +14.21).

Step 12 — houses: still not solved (at this point in the story)

Every model, on every set of houses, has an inversion cost 3–10× the human +2.63.

Model	House set	p	Upright (90.71)	Cost (+2.63)
David's*	ZuBuD, tested on all 201, 40 of them trained on	0.04	97.08 $\pm$ 0.27	+10.58 $\pm$ 0.61
r9	ZuBuD, 161 held out	0.00	92.00 $\pm$ 0.43	+10.75 $\pm$ 0.61
r8	ZuBuD, 161 held out	0.00	84.83 $\pm$ 0.60	+10.96 $\pm$ 0.64
r7	ZuBuD, all 201 trained on	0.32	90.29 $\pm$ 0.40	+11.21 $\pm$ 0.47
r8	houses_common (94, unseen by all)	0.00	86.33 $\pm$ 0.57	+22.13 $\pm$ 0.78
r7	houses_common	0.01	84.92 $\pm$ 0.55	+26.75 $\pm$ 0.83
David's	houses_common	0.00	75.75 $\pm$ 0.63	+13.13 $\pm$ 0.89

*The starred row: we test each model only on the buildings it never trained on, and for David's model that filter was quietly doing nothing — his houses are recorded under a different category name and with shorter building names than the test set uses, so it looked for matches and found none. He was tested on all 201 buildings including the 40 he had learned. We fixed the name matching and made the filter stop the run if it ever matches nothing again.

Two things to take from the table:

1. **houses_common, which we built so all three models would be on equal footing, made things worse rather than better** — costs of +13 to +27.
2. **No noise level fixes any of it.** Turning the noise up drops accuracy and cost together.

We also used to argue from this table that *training on a set of houses makes a model worse on the rest of that same set*. That rested on David's model, and it was backwards — he had trained on those buildings. Only r8 still points that way (89.2 on a set it never saw, 83.7 on held-out buildings of the set it trained on), which is one model and one comparison, so we are not claiming it.

The part I find most interesting: the *same* models, on the *same* buildings, are badly wrong on the memory task and essentially flat on the matching task. So the problem is specific to the study-then-test memory design, not to how the models represent houses — which also rules out the idea that the log-polar front end is simply orientation-sensitive for buildings, since that would break both tasks.

Steps 14 and 15 take this further, so read the above as where the house arm stood before we ran the familiarity ladders.

Step 13 — one noise level for all three categories

Until now each category got its own p, which makes the three rows of a table non-comparable. So we fitted a **single p per model** to minimise the summed distance from *both* the faces anchor (96.29) and the houses anchor (90.71) at once. Objects ride along without a vote — every object class was trained on, so objects cannot reach 84.79 at any usable noise.

Fit on houses_common — pinned at the floor (p = 0.00 / 0.01 / 0.00) for all three models, i.e. it cannot be calibrated at all. Dead end.

Fit on ZuBuD (100 seeds each):

Model	p	Faces UU / cost	Houses UU / cost	Objects UU / cost
David's*	0.26	89.25 / +14.00	90.04 / +8.38	98.83 / +3.00
r7	0.30	86.25 / +12.04	91.38 / +10.38 (<i>trained-on</i>)	98.04 / +2.46
r8	0.00	92.12 / +12.62	84.83 / +10.96	98.29 / +1.17

r8 pins at zero — no headroom, so it cannot be jointly calibrated. r7's houses are its own training buildings. **David's model is the only one that takes a real joint p**, which is why the last round focuses on it.

*His house column has the problem described in Step 12 — 40 of those buildings were ones he trained on — and because the noise level is chosen from faces and houses together, the 0.26 itself is affected. We have rerun this with the houses properly held out and are still working out which noise level to report, so his corrected numbers are not in this document yet.

Which noise level to use

Before we found the house-filter problem, we compared two candidate noise levels on this model, 0.24 and 0.26, at 100 seeds each on the same draws. Neither inversion cost differed in a way you could call real (faces -0.12 ± 0.48 , houses $+0.83 \pm 0.48$), and the total distance from the human numbers was 15.21 against 15.29 — a dead heat.

Both of those runs scored the houses wrongly, so that comparison is moot. We are now rerunning this model at several fixed noise levels with the houses properly held out, and will pick one from those.

The matching task is a separate fit, since it is a different task with its own human number, and it is unaffected by any of this. David's model on `faces_mixed64` at $p = 0.39$ reads **87.33 ± 0.35 upright** against the human 87.50, with an inversion effect of **$+12.17 \pm 0.26$** against $+10.70$ to $+11.60$.

Step 14 — in David's model, the house cost tracks familiarity

The house arm had been stuck since Step 2 — every model, every set of houses, an inversion cost $3\text{--}10\times$ the human $+2.63$, and no noise level fixing it. A familiarity ladder gives us the first real handle on it.

Everything in this step is David's model only. The ladder has been run on that one model, and as the last part of this step shows, it does not carry over to the others.

The design. Eleven 40-building sets, identical in every way except composition: k of the buildings are ones **David's model** trained on ("familiar"), the other $40-k$ are ZuBuD buildings it never saw. Only the familiar/unfamiliar ratio moves. One fixed $p=0.26$ throughout, 100 seeds per rung (`runs/sim_seeds_hladder`).

Familiar houses (of 40)	0	4	8	12	16	20	24	28	32	36	40
Unfamiliar	40	36	32	28	24	20	16	12	8	4	0
Upright-upright	92.80	92.87	93.87	92.27	92.87	96.07	96.13	97.13	97.67	98.47	100.00
Inversion cost	+10.00	+8.60	+7.93	+6.67	+5.53	+5.33	+4.60	+4.00	+3.00	+2.00	+0.40

(SEMs ± 0.16 to ± 0.86 . Human: 90.71 upright, +2.63 cost.)

For David's model the cost falls steadily as the houses get more familiar, from +10.00 to +0.40 — a 25-fold range produced by nothing but which buildings are in the set. The human +2.63 sits between $k=32$ and $k=36$, i.e. around **85% of the buildings familiar**.

This is not just David's model running out of room

The obvious objection: its upright accuracy also climbs across the ladder ($92.8 \rightarrow 100$), so the cost may be shrinking simply because there is less room to fall. The first four rungs answer it. Paired by seed, $k=0$ vs $k=16$:

	k=0	k=16	Paired difference
Upright-upright	92.80	92.87	-0.07 ± 0.81 (no real difference)
Inversion cost	+10.00	+5.53	$+4.47 \pm 1.22$ ($t=3.66$)

Accuracy is identical and the cost halves. Over that stretch familiarity is doing the work on its own, with no headroom change to hide behind. (Higher up the ladder the two do move together — $k=16 \rightarrow 32$ gains 4.8 points of accuracy alongside its 2.5-point cost drop — so the top rungs are partly compressed. The bottom half is the clean evidence.)

Faces run the same way

The face ladder points the same direction — more unfamiliar, more inversion cost:

	All familiar	All unfamiliar
Faces (r8, Step 10)	+6.92	+15.92
Houses (David's model, this step)	+0.40	+10.00

So familiarity moves the cost in both categories, though these are two different models and the house line is the only one measured rung by rung.

It is David's model that does this, not the models in general

Inside David's model, familiarity clearly drives the house inversion cost — that is what the ladder shows, and the flat-accuracy rungs make it hard to explain away.

What it does **not** explain is why the house cost stays large in the other models. r7 trained on all 201 ZuBuD buildings and was tested on those same buildings, so every house in its test was as familiar

as a house can be — and run on the *same 40-building sets at the same noise* as the ladder above, it reads **+5.47**, an order of magnitude above David’s +0.40. Familiarity alone does not put a model at the bottom of the ladder.

(r7’s headline house row is +11.21, but that is measured on all 201 buildings with Yin’s full 40-study/24-test design. The ladder sets hold 40 items, too few for that design, so the simulation scales down to 25 and 15 and is an easier task. Comparing the two directly would confuse the test pool with familiarity — +5.47 is the number to set against David’s +0.40.)

Step 15 — David’s model improves with familiarity, r8 and r9 do not

Step 14 showed David’s house inversion cost sliding from +10.00 to +0.40 as familiar buildings replace unfamiliar ones. The next question is whether that is how our models behave, or how *that* model behaves. r8 and r9 trained on the same 40 ZuBuD buildings as each other, so all three models can be run on the same sets and the only thing that changes is the model.

Same 40-building sets, same 25 study / 15 test design, same noise (0.26), 100 seeds per rung:

Familiar houses (of 40)	0	8	16	24	32	40
David’s model	+10.00	+7.93	+5.53	+4.60	+3.00	+0.40
r8	+9.07 ±0.97	+9.47 ±0.90	+9.53 ±0.90	+8.67 ±0.93	+10.47 ±0.87	+11.00 ±0.82
r9	+9.60 ±0.98	+10.80 ±0.93	+6.47 ±0.91	+7.00 ±0.79	+9.27 ±0.80	+8.80 ±0.77

Fitted slope per familiar building:

Model	slope	R ²	0 known → 40 known
David’s model	-0.2276	0.984	+10.00 → +0.40 (-9.60)
r8	+0.0421	0.519	+9.07 → +11.00 (+1.93)
r9	-0.0288	0.069	+9.60 → +8.80 (-0.80)

All three start in the same place and then only one of them moves. With every building unfamiliar they are indistinguishable — +9.07, +9.60, +10.00. By the top of the ladder David’s model is at +0.40 while the other two are still at +11.00 and +8.80. Across the ladder David’s cost drops 9.60 points; r8’s moves 1.93 the wrong way and r9’s 0.80 the right way, both inside the noise.

For David’s model the fall is close to a straight line — $\text{cost} = 9.62 - 0.217 \times (\text{familiar houses})$, $R^2 = 0.984$ across all eleven rungs, every rung within 0.61 points of it. Yin’s +2.63 falls at **32.2 of 40 familiar buildings, 81% of the set**. For r8 and r9 there is no line to fit; R^2 of 0.52 and 0.07 on a flat trend is another way of saying the cost does not depend on familiarity at all.

It is not that r8 and r9 fail to recognise the buildings. Their upright accuracy climbs across the ladder just as David’s does — r8 73.27 → 89.53, r9 79.93 → 98.53, David’s 92.80 → 100.00. All three plainly know the buildings they trained on. In David’s model that knowledge also buys orientation tolerance; in r8 and r9 it buys accuracy and nothing else.

What this means. The familiarity effect is a property of **one model**, not of the log-polar architecture and not of the memory task. That makes it a fact about training. r8 and r9 differ only in diet and behave the same here, so the diet is not it either. What is left is the thing David's model does that the others do not: houses withheld for the first two epochs and then trained at a fixed 20% of every batch, rather than grown through a six-stage class ladder. r7 fits the same picture — trained on all 201 buildings, so fully familiar with every house on these sets, and it reads +5.47, between David's +0.40 and r8's +11.00 rather than at the bottom.

The accuracy problem from Step 14 is unchanged. Where David's cost matches the human +2.63, its upright accuracy is 97.67 against a human 90.71.

Where it stands

Arm	Status	Best result
Kanwisher, faces	matched, all four models	r7 87.33/+11.09 · r8 87.13/+10.79 · r9 87.08/+11.15 · David's 87.33/+12.17
Yin, faces	matched	r8 faces_mixu16, p=0.23: 95.75/+12.17 vs human 96.29/+12.16
Yin, objects	cost matched, accuracy too high	cost +1.2 to +4.6 (human −1.92); accuracy 11–14 pts high because every object class was trained on
Yin, houses	not matched; one model behaves	David's model's cost falls with familiarity, +10.00 all-new to +0.40 all-known, passing the human +2.63 on the way (Steps 14–15). r8 and r9 on the same sets stay flat at +9 to +11, so this is one model's behaviour, not the architecture's